Marco Piña

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Research Areas

Quantitative Macroeconomics, Financial Econometrics

Education

PhD in Economics, **University of Virginia** 2023–2028 (expected)
MA in Economics, **University of Virginia** 2023–2024
MS in Economics, **University of Talca** 2018–2020

Work Experience

Research Assistant, EconTAI, University of Virginia Nov 2025–present

- Built a 175-question benchmark to test economic reasoning in six language models. Used an automated grader and question-specific rubrics to distinguish partial from complete answers. *Evaluation report*
- Developed two interactive trackers. The *BIS Speech Atlas* links policy tone scores to passages from 233 central bank speeches across 47 jurisdictions. The *AI labor market tracker* compares demand for AI skills in job postings across countries.
- Wrote four *AI for Economics notebook tutorials* on literature search, retrieval-augmented generation, model evaluation, and coordinating AI agents.

Economist, Central Bank of Chile Dec 2019–Aug 2023

- Developed, calibrated, and estimated structural models with financial frictions in the Macrofinancial Modeling Department. Built databases for economic analysis and coauthored research with senior economists.

Research Assistant, Fondecyt Regular Project No. 1180672 Mar 2019–Dec 2019

- Designed and estimated a multivariate volatility model using options data to study how option-implied volatilities inform market risk estimates.

Analyst Intern, Financial Market Commission Jan 2019–Mar 2019

- Reviewed stress-testing models for the Chilean banking system, analyzed administrative data on market and credit risk, and reported findings to the department.

Teaching

Teaching Assistant, University of Virginia Aug 2024–Dec 2025

Money and Banking (Fall 2025); Introduction to Macroeconomics (Spring 2025); Introduction to Microeconomics (Fall 2024).

Publications

Alfaro, Rodrigo and Marco Piña (2023). "Estimates of the US shadow-rate." *Latin American Journal of Central Banking* 4.1, 100080.

Jara, Alejandro and Marco Piña (2023). "Exchange rate volatility and the effectiveness of FX interventions: The case of Chile." *Latin American Journal of Central Banking* 4.2, 100086.

Herrera, Rodrigo and Marco Piña (2024). "Market risk modeling with option-implied covariances and score-driven dynamics." *The North American Journal of Economics and Finance* 72, 102136.

Working Papers

"Simple Implementable Financial Policy Rules," with M. Calani and J. Moreno. *Draft*

Work in Progress

"The Sovereign-Bank Nexus, Credit Cycles and Macroprudential Policy." *Abstract*

"The Credibility Cost of Redemption: Local Currency Debt and the Inflation Risk Premium." *Abstract*

Awards

Snavely Distinguished Second Year Paper Prize, University of Virginia	2025
Best Graduate of the Master's Program in Economics, University of Talca	2020

Research and Computational Skills

Generative AI	LLM evaluation, retrieval-augmented generation, and agentic research workflows
Developer tools	Git/GitHub, VS Code, Jupyter, SQL
Programming	Julia, Python, R
Deep learning	Flux, PyTorch
Software	Stata, MATLAB/Dynare
Financial data	Bloomberg, Eikon (Thomson Reuters)
Languages	Spanish (native), English (proficient)

Courses and Training

Machine Learning in Finance, CEMFI	Aug 2022
Introduction to Mathematical Thinking, Stanford Online	Jul 2021
Continuous Time in Macroeconomics with Machine Learning, University of Oxford	Jun 2021
Numerical Methods for Mathematics with Octave, UPValenciaX	Feb 2021
Occasionally Binding Constraints in Macroeconomics, Carleton-Bank of Canada	Jan 2021
Monetary Policy Analysis and Forecasting, IMFx	Jan 2021
Tools for Macroeconomists, University of Oxford	Aug 2020